

HQDFM Design for Manufacture(DFM) Report

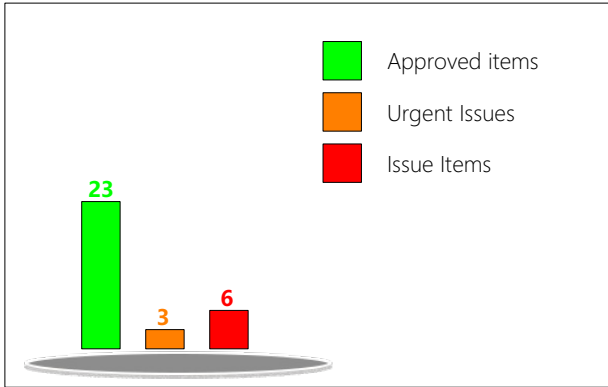
File name: 2026-06-21

Time: 2026-06-21 Layer count:4

PCB Thickness: 1.60

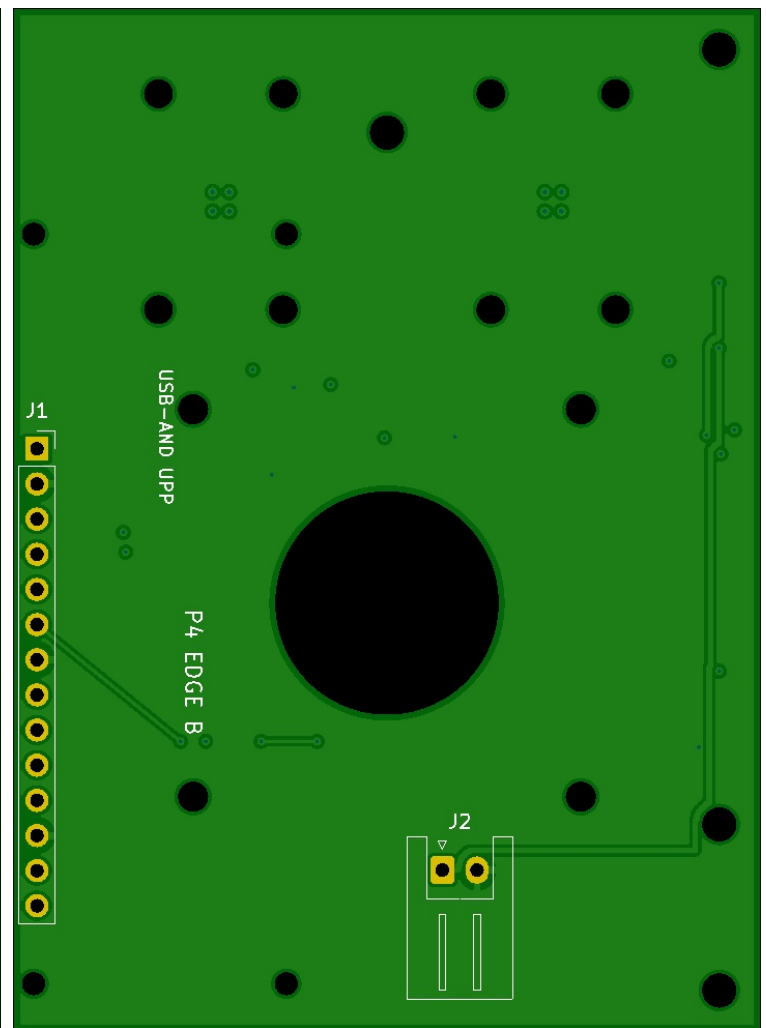
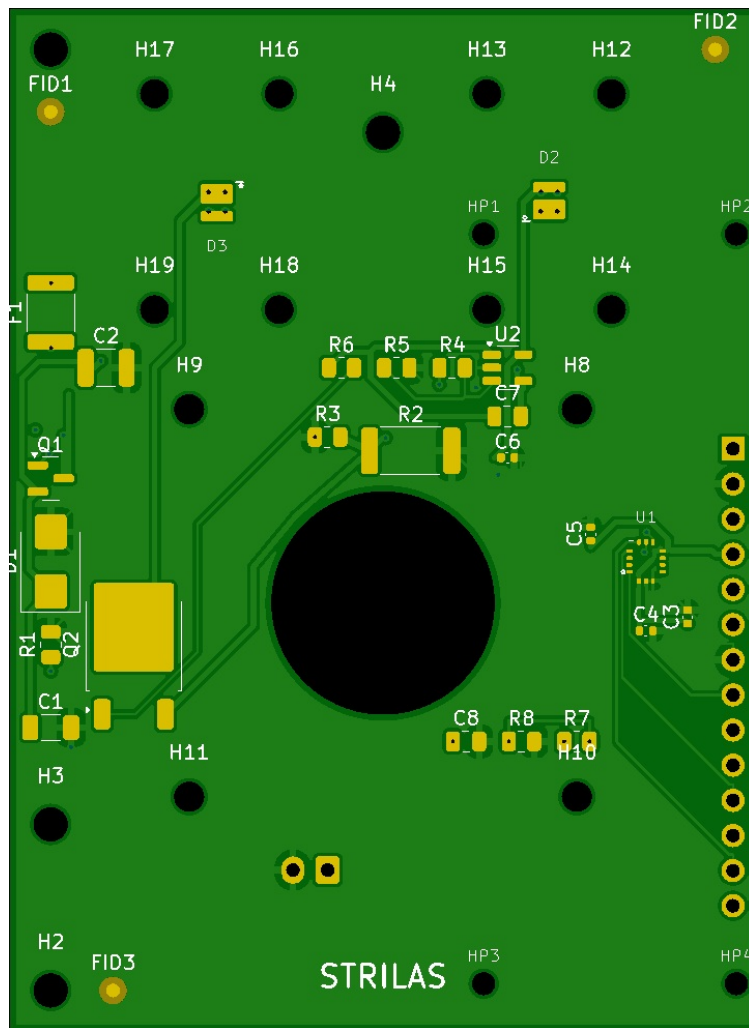
Quantity: 5

mm



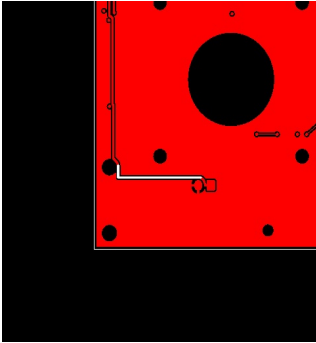
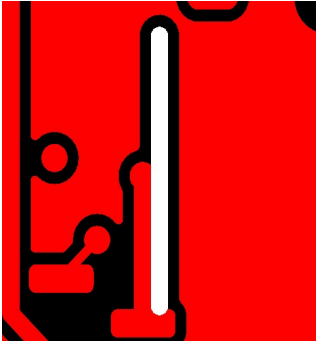
Basic Board Specs	Trace Width/Spacing	8.00/8.00mil
	Milling Density	77.4298m/m²
	Surface Finish Area	7.08%
	Test Point Count	83

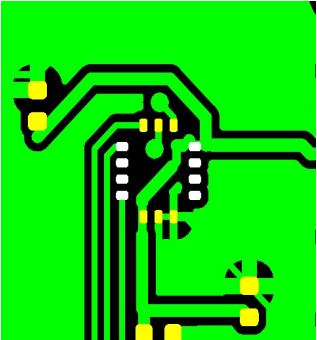
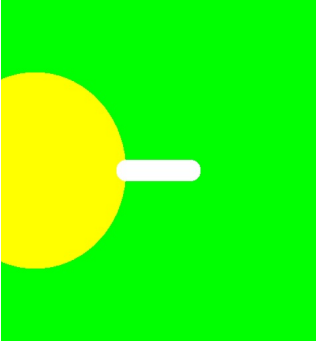
The file size is small, which can affect the surface mount assembly process. It is recommended to have a size larger than 7*7cm. You can optimize the size by adding a process edge or increasing the panelization.

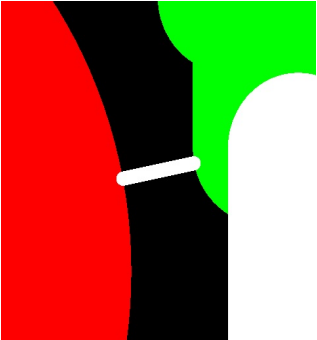
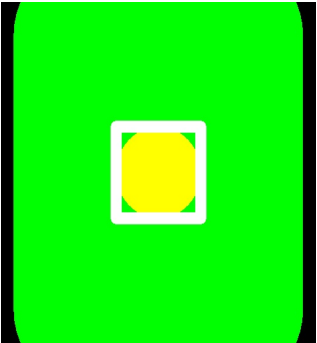


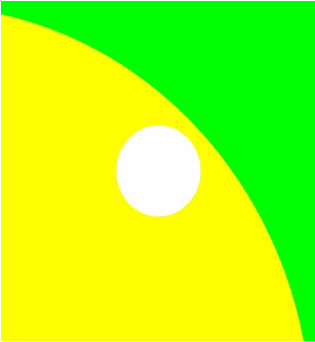
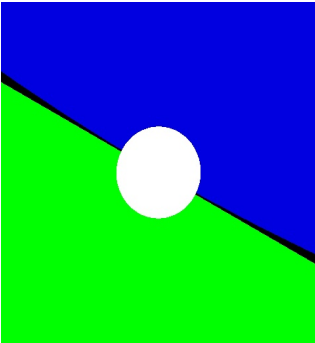
Type	Category	No. of Checks	Result
PCB Trace Analysis	Open/Shorts (IPC)	1	Fail
	Signal Integrity	4	Fail 4
	Smallest Trace Width	1	Pass 5
	Smallest Trace Spacing	3	Pass 453
	SMD Pad Spacing	1	Pass
	Pad Size	3	Pass 7 , Fail 2
	Hatched Copper Pour	2	Pass
	Annular Ring Size	2	Pass 4 , Fail 1
	Drill to Copper	5	Pass 175 , Fail 80
	Copper-to-Board Edge	2	Pass 28
	Holes on SMD Pads	4	Fail 23
PCB Drilling Analysis	Drill Diameter	8	Pass 9
	Drill Spacing	4	Fail 8
	Drill to Board Edge	4	Pass 2
	Drill Hole Density	1	Pass
	Special Drill Holes	2	Pass
	Drill Hole Errors	3	Pass
PCB Solder Mask Analysis	Solder Mask Dam	2	Pass 8 , Fail 3
	Missing SMask Opening	1	Pass
	Solder Paste Area	1	Pass
PCB Silk Analysis	Silkscreen Spacing	1	Fail 4

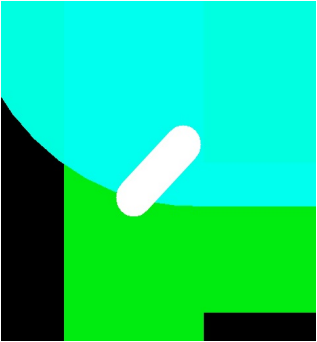
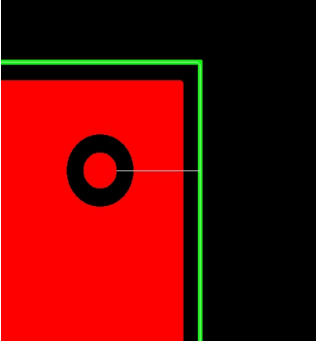
PCBA Component Analysis	Component Spacing	1	Fail
	Comp.-to-Board-Edge	3	Fail
	Component Silkscreen Spacing	0	Fail
	Pad Count Mismatch	4	Fail
	Designator Length	0	Fail
	Double-sided Components	1	Pass
	Mid Mount Comp	1	Pass
	Comp Height Analysis	2	Pass
PCBA Pin Analysis	SMT Pad Analysis	8	Fail
	Through-hole Pins	10	Fail
PCBA Pad Analysis	Chip Pads	60	Fail
PCBA Fiducial Analysis	Fiducial Count	1	Pass 3
	Fiducial Analysis	4	Pass 373 , Fail 4

ID	Check	Limits	Value	Issue	Image	Position	Qty	Level
1	Signal Integrity_Acute Angle Traces	-,-,-	Error(s) detected	There is an acute angle connection mode in your design,which will affect the signal integrity of the product.it is recommended to adjust the acute angle posintion to arc or obtuse angle connection mode		135.88,-142.77	1	Risk
2	Signal Integrity_Unconnected Traces	-,-,-	Error(s) detected	Unconnected traces or dangling lines were detected in your design. This could be the result of human error and can affect the functionality or performance of the boards. Please investigate and modify the design if necessary. Unconnected traces less than 6 mil in length should be removed.		127.31,-113.77	1	Risk

3	Pad Size_Short Pads	8,12,20	0.28 mm	Regular pads of size 0.28mm were detected in your design. This could result in overetching, detached pads and increased testing costs, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. The pads should be enlarged to at least 0.3 mm if possible.		169.00,-123.00	5	Warning
4	Annular Ring Size_Via Annular Ring	4,5,6	0.13 mm	Via annular rings of 4.92mil mil in size were detected in your design. This could increase the risk of tangency or breakouts, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended that via annular rings have a width of at least 5 mil.		137.64,-96.30	4	Warning

5	Drill to Copper_NPTH-to-Copper	8,10,12	0.21 mm	NPTH to copper spacing of 8.16mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 12 mil.		127.33,-141.73	80	Warning
6	Holes on SMD Pads_Via on SMD Pad	0.15,0.05,0	100.00 %	Holes on surface mount pads were detected in your design. During SMT assembly, solder could leak into the hole and pull solder away from the SMD contact, which could decrease manufacturing efficiency and yield, and affects the reliability of the boards. Please check and separate the holes from the pads if possible.		159.09,-136.00	23	Risk

7	Drill Spacing_Same Net Via Spacing	0.19,0.25,0.3	0.00 mm	For most factories, the same net via spacing requirement is 6-8 mil. Failure to meet the factory's requirements could increase the risk of the drill bit breaking, resulting in more drill hole defects, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. Same net via spacing of 0.00mil were detected in your design. It is recommended to increase the spacing to at least 8 mil.		137.51,-96.19	8	Risk
8	Solder Mask Dam_Solder Mask-to-Trace	1.5,2,2.5	0.00 mm	Solder mask opening to trace spacings less than 0.02mil were detected in your design. This could increase the risk of exposed traces and short circuits, which decrease manufacturing efficiency and yield. If the value is negative, exposed traces were detected in the design which are susceptible to shorts during assembly. The width should be increased to at least 0.08mm.		125.53,-91.38	3	Risk

9	Silkscreen Spacing_Solder Mask-to-Silkscreen	4,5,6	Error(s) detected	For most factories, the minimum silkscreen to solder mask spacing requirement is at least 8 mil. Failure to meet the factory's requirements could result in part of the silkscreen being removed or being printed directly on the pads, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. Silkscreen to solder mask spacing of 0 mil were detected in your design. It is recommended to increase the spacing to at least 12 mil.		162.79,-136.66	4	Risk
10	Fiducial Analysis_Fiducial-to-Board Edge	3,3.5,5	2.50 mm	The distance from the Mark point to the edge: 98.43mil were detected in your design. During SMT placement, the guide rail obstructs the Fiducial Mark point, preventing calibration. It is recommended to have a Fiducial Mark distance of ≥ 3.5mm from the board's edge		175.75,-86.00	9	Risk